exp 10

```
import cv2
import numpy as np
import matplotlib.pyplot as plt

imgL = cv2.imread("images/stereo/left.png")
imgR = cv2.imread("images/stereo/right.png")

if imgL is None or imgR is None:
print("Error: Images not found in 'images/' folder")
exit()

imgL = cv2.cvtColor(imgL, cv2.COLOR_BGR2GRAY)
imgR = cv2.cvtColor(imgR, cv2.COLOR_BGR2GRAY)

imgR = cv2.resize(imgR, (imgL.shape[1],
imgL.shape[0]))
```

```
stereo = cv2.StereoSGBM_create(
minDisparity=0,
numDisparities=16 * 5,
blockSize=5,
P1=8 * 3 * 25,
P2=32 * 3 * 25,
uniquenessRatio=10,
speckleWindowSize=100,
speckleRange=32
)

disparity = stereo.compute(imgL, imgR)

disp = cv2.normalize(disparity, None, 0, 255,
cv2.NORM_MINMAX)
disp = np.uint8(disp)

depth = np.zeros_like(disparity, dtype=np.float32)
depth[disparity > 0] = 1000 / (disparity[disparity > 0] +
1)
```

```
plt.figure(figsize=(10,4))
plt.subplot(1,3,1)
plt.title("Left Image")
plt.imshow(imgL, cmap='gray')

plt.subplot(1,3,2)
plt.title("Right Image")
plt.imshow(imgR, cmap='gray')

plt.subplot(1,3,3)
plt.title("Depth Map")
plt.imshow(disp, cmap='jet')

plt.show()

valid_depth = depth[depth > 0]
print("\n===== DEPTH SUMMARY =====")
print(f"Avg Depth : {np.mean(valid_depth):.2f}")
```

print(f"Min Depth : {np.min(valid_depth):.2f}")

print(f"Max Depth : {np.max(valid_depth):.2f}")

print("==========\n")

```
if valid_depth.size > 0:
    print(f"Avg Depth : {np.mean(valid_depth):.2f}")
    print(f"Min Depth : {np.min(valid_depth):.2f}")
    print(f"Max Depth : {np.max(valid_depth):.2f}")
else:
    print("No valid depth values found")
print("==========\n")
```

```
===== DEPTH SUMMARY =====
Avg Depth : 4.02
Min Depth : 0.79
Max Depth : 100.00
==========
```

[ ]: